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Neuron

Primer

Artificial Neural Networks for Neuroscientists: A Primer

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https://doi.org/10.1016/j.neuron.2020.09.005

SUMMARY

Artificial neural networks (ANNs) are essential tools in machine learning that have drawn increasing attention in neuroscience. Besides offering powerful techniques for data analysis, ANNs provide a new approach for neuroscientists to build models for complex behaviors, heterogeneous neural activity, and circuit connectivity, as well as to explore optimization in neural systems, in ways that traditional models are not designed for. In this pedagogical Primer, we introduce ANNs and demonstrate how they have been fruitfully deployed to study neuroscientific questions. We first discuss basic concepts and methods of ANNs. Then, with a focus on bringing this mathematical framework closer to neurobiology, we detail how to customize the analysis, structure, and learning of ANNs to better address a wide range of challenges in brain research. To help readers garner hands-on experience, this Primer is accompanied with tutorial-style code in PyTorch and Jupyter Notebook, covering major topics.

1. ARTIFICIAL NEURAL NETWORKS IN NEUROSCIENCE

Learning with artificial neural networks (ANNs), or deep learning, has emerged as a dominant framework in machine learning nowadays (LeCun et al., 2015), leading to breakthroughs across a wide range of applications, including computer vision (Krizhevsky et al., 2012), natural language processing (Devlin et al., 2018), and strategic games (Silver et al., 2017). Some key ideas in this field can be traced to brain research: supervised learning rules have their roots in the theory of training perceptrons, which, in turn, was inspired by the brain (Rosenblatt, 1962); the hierarchical architecture (Fukushima and Miyake, 1982) and convolutional principle (LeCun and Bengio, 1995) were closely linked to our knowledge about the primate visual system (Hubel and Wiesel, 1962; Felleman and Van Essen, 1991). Today, there is a continued exchange of ideas from neuroscience to the field of artificial intelligence (Hassabis et al., 2017).

At the same time, machine learning offers new and powerful tools for systems neuroscience. One utility of the deep learning framework is to analyze neuroscientific data (Figure 1). Indeed, the advances in computer vision, especially convolutional neural networks, have revolutionized image and video data processing. For instance, uncontrolled behaviors over time, such as micro-movements of animals in a laboratory experiment, can now be tracked and quantified efficiently with the help of deep neural networks (Mathis et al., 2018). Innovative neurotechnologies are producing a deluge of big data from brain connectomics, transcriptome, and neurophysiology, the analyses of which can benefit from machine learning. Examples include image segmentation to achieve detailed, micrometer scale, reconstruction of connectivity in a neural microcircuit (Januszewski et al., 2018; Helmstaedter et al., 2013), and estimation of neural firing rate from spiking data (Pandarinath et al., 2018).

This Primer will not be focused on data analysis; instead, our primary aim is to present basic concepts and methods for the development of ANN models of biological neural circuits in the field of computational neuroscience. It is noteworthy that ANNs should not be confused with neural network models in general. Mathematical models are all ''artificial'' because they are not biological. We denote by ANNs specifically models that are in part inspired by neuroscience yet for which biologically justification is not the primary concern, in contrast to other types of models that strive to be built on quantitative data from the two pillars of neuroscience: neuroanatomy and neurophysiology. The use of ANNs in neuroscience (Zipser and Andersen, 1988) and cognitive science (Cohen et al., 1990) dates back to the early days of ANNs (Rumelhart et al., 1986). In recent years, ANNs are becoming increasingly common model systems in neuroscience (Yamins and DiCarlo, 2016; Kriegeskorte, 2015; Sussillo, 2014; Barak, 2017). There are three reasons for which ANNs or deep learning models have already been, and will likely continue to be, particularly useful for neuroscientists.

First, fresh modeling approaches are needed to meet new challenges in brain research. Over the past decades, computational neuroscience has made great strides and become an integrated part of systems neuroscience (Abbott, 2008). Many insights have been gained through integration of experiments and theory, including the idea of excitation and inhibition balance (van Vreeswijk and Sompolinsky, 1996; Shu et al., 2003) and normalization (Carandini and Heeger, 2011). Progress was also made in developing models of basic cognitive functions, such as simple decision making (Gold and Shadlen, 2007; Wang, 2008). However, real-life problems can be incredibly complex;

Neuron, 107 (2020) 1048-1070. doi:10.1016/j.neuron.2020.09.005

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